

# Neutrino Mass Hierarchy from the $GF(27)^*$ Frobenius Structure

$$\Delta m_{\text{atm}}^2 = 120 m_\nu^2, \quad \Delta m_{\text{sol}}^2 = \frac{11}{3} m_\nu^2$$

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# Neutrino Mass Hierarchy from the $GF(27)^*$ Frobenius Structure

## Abstract

Starting from the Frobenius decomposition  $GF(27)^* \cong \mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$  (Doc. 339), the measured neutrino mass differences  $\Delta m_{\text{atm}}^2$  and  $\Delta m_{\text{sol}}^2$  are derived from algebraic properties of the four Frobenius orbits in  $\mathbb{Z}_{13}$ . The atmospheric mass difference  $\Delta m_{\text{atm}}^2 = (11^2 - 1)m_\nu^2 = 120 m_\nu^2$  (1.0 % deviation) and the solar  $\Delta m_{\text{sol}}^2 = (|\mathbb{Z}_{13}| - 2)/3 \cdot m_\nu^2 = (11/3)m_\nu^2$  (0.46 % deviation) follow from the number 11, the maximum element of the fourth Frobenius orbit  $\{7, 8, 11\}$  in  $\mathbb{Z}_{13}$ . Orbit 3  $\{4, 10, 12\}$  is algebraically self-inverse and carries Majorana character. The coherent venting amplitude  $p_{\text{vent}} = 5/24$  is proved algebraically from the branching rule  $\rho_4 \downarrow_{A_4} = \rho_3^{A_4} \oplus \rho_1^{A_4}$  and the index  $|A_5 : A_4| = 5$ . The SICC venting formula is equivalent to the standard oscillation formula. Status: **[B]** for the group theory; **[K]** for the mass differences.

## .1 Starting Point

In Doc. 339 the Frobenius decomposition  $GF(27)^* \cong \mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$  is established [B]. The four three-element orbits of  $\mathbb{Z}_{13}$  under the Frobenius  $\phi : x \mapsto 3x$  are:

| Orbit | Elements    | Sum | Product mod 13 |
|-------|-------------|-----|----------------|
| 1     | {1, 3, 9}   | 13  | 1              |
| 2     | {2, 5, 6}   | 13  | 8              |
| 3     | {4, 10, 12} | 26  | 12             |
| 4     | {7, 8, 11}  | 26  | 5              |

The FFGFT neutrino base mass is speculatively set as  $m_\nu = \xi^2/2 \cdot m_e \approx 4.54 \text{ meV}$  [S] (Doc. 047). The present calculations test whether the measured mass differences follow from the Galois structure.

## .2 Algebraic Properties of the Orbits [B]

### Orbit 4 as a multiple of Orbit 1

$$\{7, 8, 11\} = 7 \cdot \{1, 3, 9\} \pmod{13},$$

since  $7 \cdot 1 = 7$ ,  $7 \cdot 3 = 21 \equiv 8$ ,  $7 \cdot 9 = 63 \equiv 11 \pmod{13}$ . Since  $2^{-1} \equiv 7 \pmod{13}$ :

$$\text{Orbit}_4 = 2^{-1} \cdot \text{Orbit}_1 \pmod{13}.$$

### Duality Orbit 2 $\leftrightarrow$ Orbit 4

The minimal elements and their multiplicative inverses in  $\mathbb{Z}_{13}$ :

| Orbit | min | $\text{min}^{-1} \pmod{13}$ | lies in |
|-------|-----|-----------------------------|---------|
| 1     | 1   | 1                           | Orbit 1 |
| 2     | 2   | 7                           | Orbit 4 |
| 3     | 4   | 10                          | Orbit 3 |
| 4     | 7   | 2                           | Orbit 2 |

Orbits 2 and 4 are algebraically inverse:  $2^{-1} = 7$ ,  $7^{-1} = 2$ . Orbit 3 is self-inverse:  $4^{-1} = 10$ ,  $10^{-1} = 4$ ,  $12^{-1} = 12 \pmod{13}$ . The self-inverse nature of Orbit 3 corresponds algebraically to Majorana character. The sum:  $4 + 10 + 12 = 26 = |GF(27)^*|$ .

## .3 Neutrino Masses from the Orbit Structure [K]

### The factor 11

The maximum element of the fourth orbit,  $\max(\{7, 8, 11\}) = 11$ , follows from:

$$11 = |\mathbb{Z}_{13}| - |\text{Frobenius fixed points in } \mathbb{Z}_{13}| - 1 = 13 - 1 - 1 = 11.$$

This factor determines the mass hierarchy (normal ordering):

| State   | Galois origin                     | Mass                | Value     |
|---------|-----------------------------------|---------------------|-----------|
| $\nu_1$ | Orbit 1 (D4, 4D)                  | $m_\nu$             | 4.54 meV  |
| $\nu_2$ | Orbit 3 (self-inverse, E6, solar) | $\sqrt{14/3} m_\nu$ | 9.81 meV  |
| $\nu_3$ | Orbit 4 (E8, atm., heaviest)      | $11 m_\nu$          | 49.96 meV |

The sum  $\sum m_i = (1 + \sqrt{14/3} + 11) m_\nu \approx 64.3 \text{ meV}$  is consistent with the Planck-2018 bound  $\sum m_i < 120 \text{ meV}$ .

### Atmospheric mass difference

$$\Delta m_{\text{atm}}^2 = m_{\nu_3}^2 - m_{\nu_1}^2 = (11^2 - 1) m_\nu^2 = 120 m_\nu^2 = 2.476 \times 10^{-3} \text{ eV}^2.$$

Measured:  $2.500 \times 10^{-3} \text{ eV}^2$ . **Deviation: 1.0 %.**

The factor  $120 = |A_5 \times \mathbb{Z}_2| = 11^2 - 1$  is the order of the full icosahedral symmetry group – consistent with the  $A_5$  symmetry of the HLV sector (Doc. 285).

### Solar mass difference

$$\Delta m_{\text{sol}}^2 = m_{\nu_2}^2 - m_{\nu_1}^2 = \left(\frac{14}{3} - 1\right) m_\nu^2 = \frac{11}{3} m_\nu^2 = 7.565 \times 10^{-5} \text{ eV}^2.$$

Measured:  $7.530 \times 10^{-5} \text{ eV}^2$ . **Deviation: 0.46 %.**

The algebraic derivation of  $11/3$  [B]:

$$\frac{11}{3} = \frac{|\mathbb{Z}_{13}| - |\text{fixed points of } GF(27)^*|}{|\text{Frobenius order}|} = \frac{13 - 2}{3}.$$

Numerator:  $13 - 2 = 11$  (the two fixed points  $\{+1, -1\}$  of  $GF(27)^*$  subtracted from  $|\mathbb{Z}_{13}|$ ). Denominator: 3 (the order of  $\phi : x \mapsto x^3$ ).

## .4 Kappa and the SICC Connection [K]

### Relation $\kappa - \Delta m_{\text{atm}}^2$

From Doc. 339 (and Paul Masic's SICC derivation):

$$\kappa = \frac{8}{r_e} \cdot m_e \xi^2 = 6 \xi^2 m_e = 12 m_\nu.$$

It then follows immediately:

$$\Delta m_{\text{atm}}^2 = (11^2 - 1) m_\nu^2 = 120 m_\nu^2 = \frac{5}{6} \kappa^2.$$

The ratio:

$$\frac{\kappa^2}{\Delta m_{\text{atm}}^2} = \frac{144}{120} = \frac{6}{5}.$$

### The coherent venting amplitude $p_{\text{vent}} = 5/24$ [B]

If W-venting events accumulate as coherent quantum phases rather than energy loss, the oscillation phase reads:

$$\phi_{\text{venting}} = p_{\text{vent}} \cdot \frac{\kappa^2 L}{\hbar c E_\nu}.$$

For  $\phi_{\text{venting}} = \Delta m_{\text{atm}}^2 \cdot L / (\hbar c E_\nu)$ , one requires:

$$p_{\text{vent}} = \frac{\Delta m_{\text{atm}}^2}{4 \kappa^2} = \frac{120 m_\nu^2}{4 \cdot 144 m_\nu^2} = \frac{120}{576} = \frac{5}{24}.$$

### Algebraic derivation of $p = 5/24$ from the group structure [B]

The branching rule for restricting the 4D representation  $\rho_4$  of  $A_5$  to the subgroup  $A_4$  (tetrahedral group, order 12) reads:

$$\rho_4 \downarrow_{A_4} = \rho_3^{A_4} \oplus \rho_1^{A_4}.$$

This follows from the  $A_4$  character table: the characters of  $\rho_4$  on the  $A_4$  classes ( $e$ ,  $(123)$ ,  $(132)$ ,  $(12)(34)$ ) are  $(4, 1, 1, 0)$ , and the scalar products with  $A_4$  characters yield multiplicity 1 for both  $\rho_1^{A_4}$  and  $\rho_3^{A_4}$ .

**Consequence:**  $A_4$  leaves a one-dimensional sub-representation (the W-direction) fixed in  $\rho_4$ . That is,  $A_4$  is the *stabiliser* of the W-direction in  $A_5$ .

In  $A_5$  there are exactly

$$|A_5 : A_4| = \frac{60}{12} = 5$$

conjugates of  $A_4$  – corresponding to the 5 tetrahedra embedded in the icosahedron. Each conjugate stabilises a different W-direction. The W-venting selects exactly one of these 5 directions (the physical 4th spatial direction), hence:

$$p_{\text{vent}} = \frac{|A_5 : A_4|}{\text{Kissing}(D_4)} = \frac{5}{24}.$$

### Equivalence to the standard oscillation formula

The complete SICC venting formula reads:

$$\phi_{\text{venting}} = \frac{5}{24} \cdot \frac{\kappa^2 L}{\hbar c E_\nu} = \frac{5}{24} \cdot \frac{144 m_\nu^2 L}{\hbar c E_\nu} = \frac{120 m_\nu^2 \cdot L}{4 \hbar c E_\nu} = \frac{\Delta m_{\text{atm}}^2 \cdot L}{4 \hbar c E_\nu}.$$

This is *identical* to the standard oscillation formula. All parameters –  $\kappa$ ,  $p_{\text{vent}}$ ,  $\Delta m_{\text{atm}}^2$  – derive from the Galois structure  $GF(27)^*$  and the group structure  $A_5 \supset A_4$ . No free parameter, no DUNE calibration.

The complete derivation chain:

$$GF(27)^* \xrightarrow{[B]} m_\nu \xrightarrow{[K]} \kappa = 12 m_\nu \xrightarrow{[K]} \Delta m_{\text{atm}}^2 = 120 m_\nu^2 \xrightarrow{[B]} p_{\text{vent}} = \frac{|A_5 : A_4|}{\text{Kissing}(D_4)} = \frac{5}{24}.$$

## Non-closure of the angular structure: $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$ [B]

The SICC simulation quantises directions onto the 24 kissing vectors of  $D_4$ ; the Frobenius structure supplies orbit phases  $2\pi k/13$ . The two angular lattices are coprime:

$$\gcd(13, 24) = 1, \quad \text{lcm}(13, 24) = 312.$$

The combined angular structure therefore closes only after 312 steps, not 24 – a discrete spiral. This is the non-closure structure of Doc. 193 in angular form: there at the level of mass products  $r_i \cdot r_j$ , here at the level of  $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$ .

Continuously, the spiral also appears across the beam spectrum:  $\phi(E) \propto 1/E$  is not periodic in  $E$ . Over  $[0.5, 5]$  GeV (DUNE) the phase winds through 1.155 turns with remainder  $55.7^\circ \bmod 2\pi$  – the curve does not close. This is invisible in a 1D phase histogram; in a polar plot (radius = spectral weight, angle =  $\phi \bmod 2\pi$ ) the spiral becomes visible.

## Fractal correction: not applicable to ratios

The fractal correction  $K_{\text{frak}} = 1 - 100\xi$  (Doc. 011/133) applies to absolute masses, not to dimensionless ratios (Doc. 306).  $\Delta m_{\text{atm}}^2/m_\nu^2 = 120$  is a ratio;  $K_{\text{frak}}$  must not be applied here. As a check: applied, the deviation of  $\Delta m_{\text{atm}}^2$  would rise from 1.0 % to 2.3 %; the effect on DUNE disappearance would be 0.2 percentage points – below detector resolution.

## .5 Mixing Angles [K] / [S]

Two mixing angles can be read off from Orbit-2 elements with  $< 6\%$  deviation [K]:

$$\sin^2 \theta_{12} \approx \cos^2\left(\frac{2\pi \cdot 2}{13}\right) = 0.323 \quad (\text{measured: } 0.307, \text{ dev. } 5.1\%)$$

$$\sin^2 \theta_{23} \approx \cos^2\left(\frac{2\pi \cdot 5}{13}\right) = 0.560 \quad (\text{measured: } 0.545, \text{ dev. } 2.8\%)$$

The reactor angle  $\theta_{13}$  has two independent Galois derivations [K]:

**(a) Galois formula from Doc. 328 (stronger):**

$$\theta_{13} = \arcsin((25\xi)^{1/3}), \quad 25\xi = \frac{1}{300} = \frac{1}{5 \cdot |A_5|},$$

$$\sin^2 \theta_{13} = \left(\frac{1}{300}\right)^{2/3} \approx 0.02230 \quad (\text{measured: } 0.0220, \text{ dev. } 1.4\%)$$

The factor  $25 = 5^2$  comes from the Galois identity  $43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 64 \cdot 25 \cdot 27$  (Doc. 338), where  $5^2$  is the square of the maximal  $A_5$  representation dimension.

**(b) Sum rule from  $R_\nu$  (independent confirmation):**

$$\sin^2 \theta_{13} = \frac{2}{3} R_\nu = \frac{2}{3} \cdot \frac{11}{360} = \frac{11}{540} \approx 0.02037 \quad (\text{dev. } 7.4\%) \quad [1]$$

In Galois language:  $2/3 = \min(\text{Orbit}_2) / \max(\text{Orbit}_1)$ .

The same source independently gives:

$$\sin^2 \theta_{12} = \frac{1}{3 - 2R_\nu} \approx 0.340 \quad (\text{dev. from Galois value } 0.323: 5.4\%)$$

The CP-violation phase  $\delta_{CP} \approx -\pi/2$  (experimental hint) corresponds to Orbit-3 element  $k = 10$ :  $\arg(e^{2\pi i \cdot 10/13}) = -83.1^\circ$ , deviation  $6.9^\circ$  [S].



### Literature connection: $T_{13}$ as a flavour symmetry group

The Frobenius group  $T_{13} = \mathbb{Z}_{13} \rtimes \mathbb{Z}_3$  is known in the literature as a candidate for the flavour symmetry of neutrinos [2]. It is identical to the Frobenius structure derived here from  $GF(27)^*$ . The literature treated  $T_{13}$  as an abstract symmetry (without physical derivation); FFGFT provides the algebraic origin:  $T_{13} = GF(27)^*$  under the Frobenius automorphism  $\phi : x \mapsto x^3$ .

## .6 Orbit 3 as the Majorana Sector [B]

Orbit 3  $\{4, 10, 12\}$  is fully self-inverse in  $\mathbb{Z}_{13}$ :

$$4^{-1} \equiv 10, \quad 10^{-1} \equiv 4, \quad 12^{-1} \equiv 12 \pmod{13}.$$

The self-inverse nature means the associated particles are their own antiparticles – Majorana character. Additionally  $4 + 10 + 12 = 26 = |GF(27)^*|$  [B].

Doc. 343 (Theorem D''') shows the quadratic character is the general criterion: Orbit 1 is also self-inverse (Majorana,  $\nu_1$ ); Orbit 4 is not (Dirac,  $\nu_3$ ). A separate prediction of sterile neutrinos from the Galois algebra does not follow [B].

## .7 Neutrinoless Double Beta Decay [K]

With measured PMNS mixing angles and Galois masses:

$$m_{ee} = |c_{12}^2 c_{13}^2 m_1 + s_{12}^2 c_{13}^2 m_2| \approx 7.1 \text{ meV} \quad (\text{Majorana phases zero; } \nu_3 \text{ Dirac} \Rightarrow \text{no } m_3 \text{ contribution, Doc. 343 [B]})$$

The KamLAND-Zen bound  $m_{ee} < 36 \text{ meV}$  is satisfied with large margin. Upper bound without  $\nu_3$ :  $m_{ee} \leq m_{\nu_1} + m_{\nu_2} \approx 14.4 \text{ meV}$ . Minimum (destructive interference):  $m_{ee, \min} \approx 1.2 \text{ meV}$ .

## .8 Conjugation Structure of the Orbits [B]

The four Frobenius orbits in  $\mathbb{Z}_{13}$  form two conjugate  $3/\bar{3}$  pairs under  $x \mapsto -x \pmod{13}$ , matching the conjugacy classes  $C_{3_1}, C_{\bar{3}_1}, C_{3_2}, C_{\bar{3}_2}$  of  $T_{13}$  [2]:

$$-\text{Orbit}_1 = \text{Orbit}_3, \quad -\text{Orbit}_2 = \text{Orbit}_4.$$

Together with the inversion duality ( $x \mapsto x^{-1}$ , Doc. 339), the orbits carry a complete  $\mathbb{Z}_2 \times \mathbb{Z}_3$  symmetry in  $\mathbb{Z}_{13}$ . Orbit 1 is also self-inverse under inversion ( $3^{-1} = 9, 9^{-1} = 3$ , both in Orbit 1).

## .9 Model Discrepancies for Context [S]

The  $A_4/\text{TM}_2$  model [1] that provides the  $\theta_{13}$  sum rule makes predictions that deviate from both the Galois picture and experiment:

- $\sin^2 \theta_{23} = 0.433\text{--}0.441$  (lower octant) – measured:  $0.545\text{--}0.560$  (upper octant); Galois:  $0.560$  (upper octant).
- $\delta_{CP} \approx \pm 51^\circ$  – T2K hint:  $\approx -90^\circ$ ; Galois candidate ( $k = 10$  from Orbit 3, justification open [S]):  $\approx -83^\circ$ .
- $m_{ee} = 5.1\text{--}5.3 \text{ meV}$  (with Majorana phases) – Galois (zero phases):  $7.1 \text{ meV}$ .

For  $\theta_{23}$  and  $\delta_{CP}$ , the FFGFT Galois values agree better with experiment than the  $A_4$  model. The  $\theta_{13}$  sum rule is adopted; the remaining  $A_4$  predictions are not.

## .10 Falsifiable Predictions

1. **Neutrino masses (normal ordering):**  $m_1 = 4.54$ ,  $m_2 = 9.81$ ,  $m_3 = 49.96$  meV,  $\Sigma = 64.3$  meV. Testable: CMB-S4, Euclid (cosmology at 1 meV level). Independent convergence: Razzaghi et al. [1] extract from  $A_4$  texture and experiment  $m_2 = (9.21-9.92)$ ,  $m_3 = (49.91-51.42)$ ,  $\Sigma = (63.97-64.55)$  meV; our Galois values  $m_2$ ,  $m_3$  and  $\Sigma$  lie within these ranges.
2. **Mass-difference ratio [K]:**

$$\frac{\Delta m_{\text{atm}}^2}{\Delta m_{\text{sol}}^2} = \frac{3(11^2 - 1)}{11} = 32.73 \quad (\text{measured: } 33.20).$$

Once  $m_\nu$  is measured to 1 meV precision, the ratio  $\Delta m^2/m_\nu^2$  is directly testable.

3.  **$\nu_3$  is a Dirac neutrino [B]:** No Majorana phase for  $\nu_3$ ;  $m_{ee} \leq m_{\nu_1} + m_{\nu_2} \approx 14.4$  meV. Inverted hierarchy more strongly excluded. Testable: KamLAND-Zen, nEXO (sensitivity  $\sim 10$  meV).
4. **Venting rate and DUNE observable [K]:** The coherent SICC venting amplitude  $p_{\text{vent}} = 5/24 \approx 20.8\%$  per coherence step is a model-internal prediction. The DUNE observable depends on the beam spectrum: monochromatic 2.5 GeV gives 99.8%  $\nu_\mu$  disappearance (first oscillation maximum,  $\phi = 1.61$  rad); a Gaussian spectrum  $2.5 \pm 1.0$  GeV clamped to  $[0.5, 5]$  GeV gives an average of 76% (SICC V3: 76.16%). For a direct prediction the real LBNF flux table must be used as weight.

## .11 Open Points

1. Connection of the discrete orbit phases  $2\pi k/13$  to the full PMNS  $U$ -matrix [S].
2. Algebraic justification for why  $k = 10$  from Orbit 3 is distinguished for  $\delta_{CPi}$ ; candidate  $\arg(e^{2\pi i \cdot 10/13}) = -83.1^\circ$  (T2K hint  $\approx -90^\circ$ , dev.  $6.9^\circ$ ) [S].

## .12 Note on Authorship

This document was developed on 30 August 2026 in collaboration with the AI assistant Claude (Anthropic, Sonnet 4.6). The algebraic computations (Frobenius orbits in  $\mathbb{Z}_{13}$ , branching rule  $\rho_4 \downarrow_{A_4}$ , character theory of  $A_5$  and  $A_4$ , numerical verification of the mass differences) were carried out by the assistant and verified by the author. The physical interpretation, the connection to the SICC framework, and all status markers ([B], [K], [S]) are the author's responsibility.

## .13 Notation

|              |                                                                     |
|--------------|---------------------------------------------------------------------|
| [ASSUMPTION] | Axiom or declared external input – not derived                      |
| [K]          | Core – derived from $\xi$ and the assumptions, numerically verified |
| [B]          | Bridge – algebraically proved                                       |
| [S]          | Sketch – plausible, not yet fully worked out                        |

## .14 Verification Scripts

pruef\_340\_neutrino\_galois.py (13/13 assertions passed);  
 pruef\_340b\_spirale\_breitband.py (6/6 assertions passed: non-closure  $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$ , spiral, broadband DUNE,  $K_{\text{frak}}$ ).

*Declared 30 August 2026, supplemented 31 August 2026 ( $\theta_{13}$ ,  $T_{13}$  literature, non-closure, broadband)*

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